

PROJECT ATLAS

ATLAS-LOG-002: PROGRESS SNAPSHOT & ARCHITECTURE BASELINE

Date: July 1, 2026 | Prepared for: Peter Lochhead Hepburn

Status: **PHASE 0 (STABLE)**

1. Executive Summary

The Master Flightline Clock system (Project Atlas) has transitioned from a basic timer script into a robust, database-driven control application. The core engine is finalized, the display layout is hardware-aligned for LED panels, and data integrity is ensured via SQLite persistence.

2. Current System Capabilities

Feature	Implementation Details
Persistence	SQLite3 Database handles all heat data. Survives restarts and power cycles.
Timeline Engine	10Hz pulse; manages transitions between Track Wait, Track Entry, and Heat Running phases.
Display Layout	6-Row vertical stack optimized for P10/P5 LED Matrix legibility.
GAMP Validation	Input shielding prevents invalid time windows and duplicate heat IDs.
Operator UI	Tabbed interface with integrated Daily Schedule table and Live Control monitor.

3. Key Architectural Decisions

- The Primary Timer:** A single, high-magnitude countdown that automatically changes context (e.g., from "Time to Open" to "Window Remaining").

- **State Immutability:** The Engine broadcasts a data "model" every 100ms; the UI is a "dumb" renderer that simply paints what it is told.
- **Audit Trail:** Cancelled or Completed heats are preserved in the DB with statuses for future reporting and search.

4. Roadmap (Post-Return)

- **Hardware Integration:** HDMI-to-LED configuration and HUB75 testing.
- **Configuration Management:** Database-driven settings (LED brightness, competition name).
- **Reporting:** Exporting daily heat logs to CSV/PDF.